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AI Engineer | Full-Stack Developer

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Profile

Computer Science graduate who builds applied AI and production business software end to end. Creator of SKY, a bilingual (English, Urdu, Roman Urdu) voice agent that operates a Windows PC on a single 8 GB GPU with an offline fallback, and of four tested Flask business systems, one running multi-tenant in the cloud. Brings leadership experience as co-founder of TriTec Software House and as a sales manager, and wants to apply both to AI engineering and product roles.

Education

2022 – 2026	Iqra University, Karachi	BS Computer Science
2021 – 2022	Govt. Boys Degree College	FSc Pre-Engineering
2019 – 2020	Iqra Hadiqa Tul Atfal School	Matriculation

Experience

Present — **AI Engineer & Business Software Developer** — Independent / Freelance

- Design and build AI agents that combine LLMs, NLP, speech and vision models with tool automation to run real tasks on a computer.
- Deliver full-stack billing, POS and fee systems to businesses, from database design to cloud deployment and automated testing.
- Choose models by measuring them on target hardware; tune prompts, quantisation and fallbacks for speed and reliability.

2023 – 2024 — **Co-Founder & Manager** — TriTec Software House

- Co-founded and led a software venture, growing the team and delivering projects for international
- clients. Managed operations, Fiverr and Discord client projects alongside full-time studies.

2024 — **Project Manager Intern** — SNJ Global services

- Supported project planning, tracking and cross-team coordination with project management tools. Built
- reporting dashboards, maintained documentation and automated routine workflow steps.

2024 — **Sales Manager** — SNJ Globals services

- Led international sales and project execution, building client relationships and closing deals.

Projects

2022 – 2026 **SKY — Bilingual Desktop AI Agent** Final Year Project

- Sole designer and developer of a voice agent that wakes on “Hey Sky”, understands English, Urdu and Roman Urdu, and operates Windows through 14 tool modules: WhatsApp calls, Gmail, Spotify and YouTube, files, system settings, apps and games.
- Pipeline of Silero VAD, Whisper large-v3-turbo (int8), a fuzzy command resolver and Gemini 2.5 Flash, with automatic fallback to local Llama 3.2 3B when offline. About 15,400 lines of Python and TypeScript.
- Fit 11 models into an 8 GB VRAM budget on one RTX 5070; replaced Qwen 2.5 7B (11–20 s per reply) with Llama 3.2 3B (2–3 s) after benchmarking.
- Added on-screen vision (YOLOv8n, EasyOCR), a JSON-repair parser with Pydantic v2 validation, remote commands by email, and a FastAPI, JWT and WebSocket backend with a live React 19 interface.

Stack: Python, PyTorch, CUDA, faster-whisper, Ollama, Gemini API, Ultralytics, FastAPI, SQLite, React, TypeScript · *Link:* ahsanharoon24.github.io

2026 **Business Management Software Suite** 4 systems · 200+ tests

- Restaurant POS (live): redesigned from one shared database to a separate SQLite database per restaurant; deployed on DigitalOcean with Gunicorn, Nginx, systemd and nightly backups.
- Cloud Billing SaaS: company self-signup in about two minutes, role-based staff logins, per-company invoice numbering and GST/VAT, with tested tenant isolation (53 checks).
- Invoice & Billing (delivered to client): accounting-safe invoicing, partial payments, audit log and daily backups; 41 automated checks plus 116 UAT cases.
- School Fee Management: one-click monthly challans that skip duplicates, defaulter and class-wise reports, printable A4 challans (39 checks).

Stack: Python, Flask, SQLite, multi-tenancy, role-based access, Gunicorn, Nginx, DigitalOcean
Link: ahsanharoon24.github.io/POS.html

Skills

AI & ML	LLM agents, NLP, prompt engineering, Whisper, Silero VAD, Gemini API, Ollama (Llama, Qwen), YOLOv8, EasyOCR, PyTorch, CUDA
Backend	Python, Flask, FastAPI, Pydantic v2, SQLite, multi-tenant database design, JWT authentication, WebSockets
Frontend	React 19, TypeScript, Vite, responsive HTML/CSS, mobile-first and A4 print layouts
DevOps & Testing	Linux (Ubuntu), Nginx, Gunicorn, systemd, DigitalOcean, Playwright, automated end-to- end test suites
Programming Languages	Python, TypeScript, Java, C++
	English (professional), Urdu (native)

Core Strengths

- Leadership
- Sales & clients
- Problem solving
- Automation
- Project management
- Entrepreneurship
- Data analysis
- Digital marketing

Research Interest

Reliable AI agents on consumer hardware for speakers of low-resource and code-switched languages such as Urdu and Roman Urdu.